Project Title: AI-Powered Ticket Creation and Categorization from User Input

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Project Statement:

This project is designed to automate the initial phase of ticket management by developing an AI system that can understand and process natural language requests from users. The system will automatically generate structured service tickets—including a title, description, category, and initial priority—from unstructured user input provided via chat, email, or a web form. This automation will streamline support workflows, reduce manual effort, and improve the efficiency of help desk operations.

Outcomes

- **Automated Ticket Creation:** The system will automatically generate well-structured tickets from user-provided text, eliminating the need for manual data entry.
- **Accurate Categorization:** The AI model will categorize tickets into predefined types (e.g., "Hardware Issue," "Software Bug") with a high degree of accuracy.
- **Enhanced Data Extraction:** The system will identify and extract key entities like user details, affected services, and error messages from free-form text.
- **Improved Support Efficiency:** By automating the initial triage, the project will reduce ticket handling time and allow support staff to focus on resolving issues.

Modules to be Implemented

- **Module 1: Data Collection and Preprocessing**
    - Collect a diverse dataset of past user issues and their corresponding structured ticket data.
    - Clean and preprocess the text data to handle inconsistencies, spelling errors, and irrelevant information.
    - Annotate the dataset with labels for ticket categories, priorities, and named entities.
- **Module 2: NLP Model Development**
    - Develop and train a Natural Language Processing (NLP) model for text classification to categorize tickets.
    - Implement Named Entity Recognition (NER) to extract critical information like usernames, machine names, and error codes.
    - Fine-tune the model to understand the user's intent and sentiment to assign an appropriate priority.
- **Module 3: Ticket Generation Engine**
    - Build a core engine that takes preprocessed user input and uses the trained models to generate a structured JSON object representing the ticket.
    - Map the extracted entities and classification results to specific fields such as title, description, category, and priority.

- **Module 4: User Interface and Integration Layer**
    - Develop a simple web-based interface (e.g., a chatbot or a form) for users to submit their issues.
    - Integrate the AI engine with this interface to process user input in real time.
    - Create a display to show the automatically generated ticket details to the user for confirmation.

Milestones

- **Milestone 1 : Data Preparation & Annotation**
    - Objective: Collect and prepare a clean, annotated dataset.
    - Tasks: Collect a diverse set of historical ticket data; clean and normalize text; manually annotate a portion of the data for training.
- **Milestone 2 : Core Model Development**
    - Objective: Train and validate the NLP and NER models.
    - Tasks: Select appropriate models (e.g., scikit-learn classifiers, SpaCy for NER); train the models on the annotated dataset; evaluate initial model accuracy.
- **Milestone 3 : Ticket Generation & Logic Implementation**
    - Objective: Build the ticket generation engine and business logic.
    - Tasks: Develop the core script to process input and generate ticket data; write logic to determine priority based on content and keywords; test the full end-to-end flow.
- **Milestone 4 : Deployment & Reporting**
    - Objective: Deploy the system and finalize documentation.
    - Tasks: Integrate the engine with a simple web interface; conduct final testing for performance and accuracy; prepare a report on the project's outcomes and model performance.